(Group) Coursework No. 2: Renewable Energy Balance Sheet (Excel)

1. Background

The chairman of CleanEnergy Ltd has commissioned a financial analysis into a potential bid for government funding to set up and operate a wind farm in the small island country of Windonia. The aim is to compare the profitability of different investment scenarios that depend on a number of key factors related to location and size of turbine to be used. The location of the wind farm can be *onshore* or *offshore* of Windonia and the company can choose from various sizes of wind turbines that are readily available from their main supplier (see below).

Thus, they wish to secure funding from the Central Electricity Generating Council (CEGC) of Windonia that has recently started a renewable energy investment scheme in order to replace the country's outdated infrastructure of fossil fuel power plants. Investors can bid for funding in the form of corporate borrowings for a flat annual rate of interest. In addition, the government is willing to provide an initial subsidy towards the installation costs.

Note the key characteristics and operational aspects of the wind turbines include the following:

1) Average (annual) output yield:[1] y

   This is a ratio that is calculated based on average (annual) wind speed (w):

$$y = -0.323 + 0.147w - 0.01w^2 + 0.0002w^3 \quad for \quad 20 \leq w \leq 40 \ mph$$

2) Nominal energy output: N

   The turbines are able to provide a standard energy output in kWh (kilowatt-hour) in function of the swept area of the rotor blades and the output yield (y), given by the following formula:

$$N = \left(\frac{d}{2}\right)^2 \times \pi \times y$$

   where d is the diameter of the rotor blades in meters.

3) Annual leasing costs:

   CleanEnergy Ltd will lease out from CEGC the required area for a fixed period of electricity production. In addition, they will require one extra year at the start and at the end of this period for installation and decommission of the plant, respectively (i.e. +2 years). The leasing fee for the required area will be payable in arrears as a sum of a base rental price and a royalty fee (i.e. commission) on the gross revenue from electricity sale (no commission is payable in the first and in the last year of the plant's operation).

[1] **Note**: This is different than the annual rate of return (yield) on an investment project!

4) Installation and decommission costs:

- CleanEnergy Ltd has negotiated with their main supplier company a basic price formula that depends on the rotor dimension (d), but is subject to adjustments depending on the location of the wind farm, which is specified as a cost factor (see section 2.1).
- Also, CleanEnergy Ltd will have a contractual obligation to decommission the wind turbines and to fully regenerate the used areas within one year after the end of the leasing period. The decommission costs will incur at the end of the additional year and will need to be covered entirely from company funds resulting from electricity sale during the time of the contract.
- The time needed for installation and decommission of the entire plant is estimated as 1 year each (irrespective of the size of the wind turbine).

5) Estimated annual maintenance costs:

The total costs of the (regular) annual maintenance are expressed as a percentage of the gross revenue from electricity sale in function of turbine dimensions (d) and length of operation in whole years (see Table 1).[2]

2. Excel Spreadsheet Model

Your job as an Excel expert and financial analyst advising the company is to construct a spreadsheet model of the investment project balance sheet with the following (minimum) specifications:

2.1 The INPUT parameters to the program are given as follows:

- ***Duration*** of turbine operation (T) in whole years (between 10 to 40 years)
- ***Diameter*** of the rotor blades (d) in meters (between 15m to 250m, integer by a step value of 5m)
- Average (annual) ***wind speed*** (w) in miles per hour (between 20mph to 40mph)
- Annual base ***leasing price*** (P_L) per acre payable in arrears (between £10,000 to £25,000) for the required installation area.[3]
    - The leasing area for which the rent is payable is calculated as:

$$A = 1.5 \times d^2 \ (m^2)$$

- Lease ***commission rate*** ($R_L\%$) payable as a percentage of the gross electricity sale revenue (between 10% to 20%).
- Location ***cost factor***: (k) that multiplies the standard installation and decommission costs (non-integer, between 1 to 3).
    - The installation costs cover typical initial expenses such as: purchase, delivery, infrastructure, etc., and is estimated using a base formula of:

[2] See "*Maintenance*" WS of "AS1104_T1_2024-25 - CW2-GR-.xlsx" Excel WB. [3] Note: 1 acre = 4046.86 m^2.

$$P_I = k \times e^{0.02d} \ (£\,millions)$$

- The decommission costs cover all of the final expenses such as: demolition, removal, clean-up, environmental regeneration, etc., and it is estimated as:

$$P_D = 0.4 \times P_I$$

- Initial electricity ***sale price*** (E_0) in £/kWh in the first year (between £0.15 to £0.20)
- Electricity sale price ***annual inflation rate*** $(j\%)$ applied at the end of each year (between 2.5% to 5%):
  - The electricity sale price applied in year t results as:

$$E_t = E_0 \times (1 + j)^{t-1}$$

- Installation ***location***:
  - **Onshore:** Unadjusted costs
  - **Offshore:** the base leasing costs are *reduced* by 90%, but both the installation and the decommission costs are *increased* by 30%.
- Annual ***borrowing rate*** $(i\%)$ compound effective rate on negative ending balances (between 6% to 10%) – i.e. applicable to all years $t \leq DPP$ (see below section 2.2 ii)
- Annual ***lending rate*** $(i\%)$ compound effective rate on positive ending balances (between 1% to 3%) – i.e. applicable to all years $t > DPP$ (see below section 2.2 ii)
- Subsidy ***incentive*** $(S\%)$ applied as a lump payment to cover a fraction of the total installation costs (between 15% to 25%).

2.2 The spreadsheet model OUTPUT should cover the following:[4]

i. A balance sheet (cash flow table) of the investment costs and annual returns for a single turbine (i.e. for the exact number of years specified in the inputs), which should contain (at least) the following cash flow values (e.g. column headings):

   (a) The installation costs at the outset;

   (b) The annual gross revenue from electricity sale;

   (c) The annual maintenance and leasing costs;

   (d) The annual net revenue;

   (e) The present value of the end-of-year balance (surplus) given by the cumulative net revenue up to the current year less the initial installation costs;

   (f) The decommission costs in the final year and the present value of the residual balance (profit).

ii. Based on the balance sheet from *part i.*, the model should also be able to output the following:

   (a) The annualised *discounted return on investment* (AROI) as:

$$-\frac{PV_{profit} - P_0}{P_0} \times \frac{1}{T+2} \times 100\ (\%)$$

[4] See on Moodle numerical examples and additional explanatory video about the calculation of the cash flow components of the balance sheet and model outputs.

where PV_{profit} is the present value of the final profit; and

$P_0 < 0$ is the net borrowing cost (i.e. after the subsidy).

(b) The *discounted payback period* (DPP) of the investment project, given as the length of time the initial investment is amortised by the cumulative net revenue (i.e. the end-of-year balance becomes first time positive).

Note: the input and output of the model could be on the same worksheet.

3. Coursework Tasks

Therefore, you need to build a spreadsheet model that will aid the pricing and analysis of an appropriate wind turbine investment. Thus, you should design a model that will output the exact balance sheet (i.e. for the given number of years only). Further, the model should only generate an output if all the parameters are inputted correctly.

In addition, you need to provide a detailed written description of how your model works. Think of it as a detailed ***user guide*** that you need to provide to a fellow colleague. So, the document should describe the layout and workings of the model, any error traps or aspects the user need to be aware of, list and explain the Excel formulas to create the output, etc.

Other than that, you might describe here any additional features that you have managed to implement in the model (e.g. other than the minimum specifications, such as extra graphical illustrations of the balance sheet, cash flow, analysis of optimal turbine size, etc.).[5]

The format and presentation of the user guide is left to the imagination and creativity of each group. The minimum requirement is that it should be well presented, easy to follow/read, have a clear structure (i.e. sections and subsections) and be between 3-4 pages long (including any figures/graphs, tables that you might want to add, use any reasonable type 11pt font of your choice).

(Tentative) Marking Scheme[6]

Report (User Guide) (*.docx file)	Excel Spreadsheet Model (*.xlsx file)				Bonus
	Model Structure (Input/Output)	Balance Sheet Table	Intermediate & Summary	Quality & Efficiency of Implementation	
30	10	40	10	10	5

[5] Of course, for some appropriate bonus marks.
[6] Minor changes are possible.

4. Submission

Submit on Moodle the user guide document (.docx) and a final and clean Excel workbook (.xlsx) containing your data and spreadsheet model. There should be only a **single** submission by each group before the deadline of the first coursework (see CAP form)! Note that any submission after this deadline will be subject to a 10% penalty per day on the coursework grade!

Don't forget that this is a peer-assessment (PA) type coursework whereas everyone needs to rate their teammates. The PA should be honest and should reflect on aspects such as: availability, organisation, co-operation with the rest of the team, creativity, meeting deadlines and competency in their roles (hence keeping a good track record of the group work will be helpful). 'No contribution' type feedback will be scrutinised and will only be accepted when there are genuine reasons for them.